

Experiment # 08

Implementation of Color Models and Color Image Processing

Objective

1. To implement color image processing

Software

- Python Idle 3.7 or 3.8.5
- Pycharm

Theory

The human visual system can distinguish hundreds of thousands of different colour shades and intensities, but only around 100 shades of grey. Therefore, in an image, a great deal of extra information may be contained in the colour, and this extra information can then be used to simplify image analysis, e.g. object identification and extraction based on colour.

Three independent quantities are used to describe any particular colour. The *hue* is determined by the dominant wavelength. Visible colours occur between about 400nm (violet) and 700nm (red) on the electromagnetic spectrum, as shown in given figure 1.

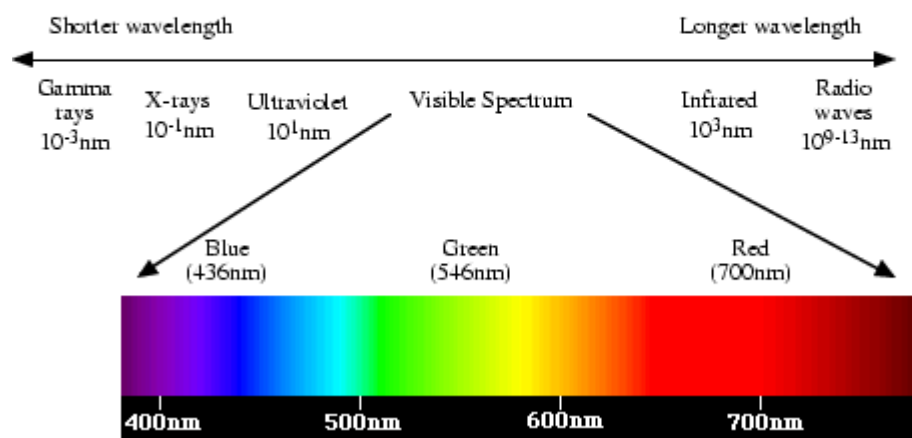


Figure 1: The visible spectrum.

The *saturation* is determined by the excitation purity, and depends on the amount of white light mixed with the hue. A pure hue is fully saturated, i.e. no white light mixed in. Hue and saturation together determine the *chromaticity* for a given colour. Finally, the *intensity* is determined by the actual amount of light, with more light corresponding to more intense colours.

Achromatic light has no colour - its only attribute is quantity or intensity. Greylevel is a measure of intensity. The *intensity* is determined by the energy, and is therefore a physical quantity. On the other hand, *brightness* or *luminance* is determined by the perception of the colour, and is therefore psychological. Given equally intense blue and green, the blue is perceived as much darker than the green. Note also that our perception of intensity is nonlinear, with changes of normalised intensity from 0.1 to 0.11 and from 0.5 to 0.55 being perceived as equal changes in brightness.

In the RGB model, an image consists of three independent image planes, one in each of the primary colours: red, green and blue. Specifying a particular colour is by specifying the amount of each of the primary components present. Figure 2 shows the geometry of the RGB colour model for specifying colours using a Cartesian coordinate system. The greyscale spectrum, i.e. those colours made from equal amounts of each primary, lies on the line joining the black and white vertices.

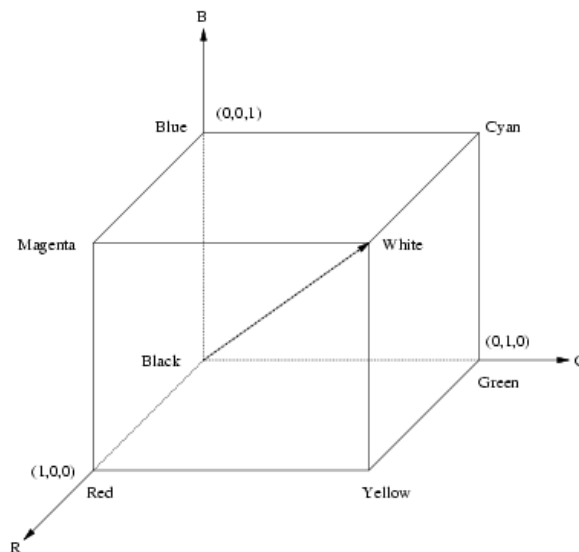


Figure 2: The RGB colour cube. The greyscale spectrum lies on the line joining the black and white vertices.

This is an *additive* model, i.e. the colours present in the light add to form new colours, and is appropriate for the mixing of coloured light for example. The image on the left of figure 3 shows the additive mixing of red, green and blue primaries to form the three secondary colours yellow (red + green), cyan (blue + green) and magenta (red + blue), and white ((red + green + blue).

The RGB model is used for colour monitors and most video cameras.

The CMY Model

The CMY (cyan-magenta-yellow) model is a *subtractive* model appropriate to absorption of colours, for example due to pigments in paints. Whereas the RGB model asks what is added to black to get a particular colour, the CMY model asks what is subtracted from white. In this case,

the primaries are cyan, magenta and yellow, with red, green and blue as secondary colours (see the image on the right of figure 3).

When a surface coated with cyan pigment is illuminated by white light, no red light is reflected, and similarly for magenta and green, and yellow and blue. The relationship between the RGB and CMY models is given by:

$$\begin{aligned} C &= 1 - R \\ M &= 1 - G \\ Y &= 1 - B \end{aligned}$$

The CMY model is used by printing devices and filters.

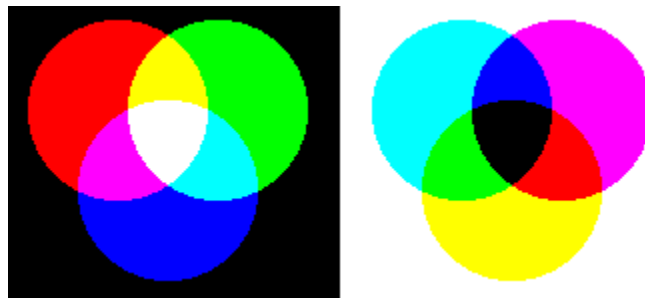


Figure 3: Mixing of colors

The figure on the left shows the additive mixing of red, green and blue primaries to form the three secondary colours yellow (red + green), cyan (blue + green) and magenta (red + blue), and white ((red + green + blue)). The figure on the right shows the three subtractive primaries, and their pairwise combinations to form red, green and blue, and finally black by subtracting all three primaries from white.

The HSI Model

As mentioned above, colour may be specified by the three quantities hue, saturation and intensity. This is the HSI model, and the entire space of colours that may be specified in this way is shown in figure 4, showing the HSI solid on the left, and the HSI triangle on the right, formed by taking a horizontal slice through the HSI solid at a particular intensity. Hue is measured from

red, and saturation is given by distance from the axis. Colours on the surface of the solid are fully saturated, i.e. pure colours, and the greyscale spectrum is on the axis of the solid. For these colours, hue is undefined.

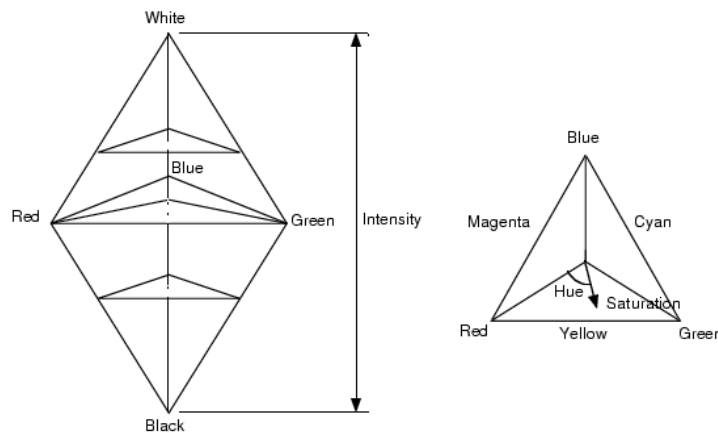


Figure 7: The HSI model

Conversion between the RGB model and the HSI model is quite complicated. The intensity is given by

$$I = \frac{R+G+B}{3},$$

where the quantities R, G and B are the amounts of the red, green and blue components, normalised to the range [0,1]. The intensity is therefore just the average of the red, green and blue components. The saturation is given by:

$$S = 1 - \frac{\text{Min}(R,G,B)}{I} = 1 - \frac{\text{min}(R,G,B)}{R+G+B}$$

where the min(R,G,B) term is really just indicating the amount of white present. If any of R, G or B are zero, there is no white and we have a pure colour.

Some Useful Commands:

1. OpenCV reads image in BGR format by default
cv2.cvtColor(nemo, cv2.COLOR_BGR2RGB)
2. RGB to HSV conversion: cv2.cvtColor(nemo, cv2.COLOR_RGB2HSV)
3. Binary threshold: cv2.inRange(Image, lower_range, upper_range)

4. Pseudo coloring of gray image: `cv2.applyColorMap(img, cv2.COLORMAP_JET)`

Lab Task:

1. Convert CMY to RGB with key color 0.412 where the formula is
 $[R \ G \ B] = 255 * [1 - C \ 1 - M \ 1 - Y] * [1 - key \ 1 - key \ 1 - key]$
2. Extract blue color letters from the given image where the color range is (100,0,0) to (130,255,255)

Marketing

3. Apply HSV and RAINBOW color map to the gray scale image